

Tugce Aldemir

Assistant Professor, Department of Teaching, Learning, and Culture
Collaborative Human-AI STEM Learning Ecologies in STEM+C (CoHALE) Lab
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Research Program

My research program is grounded in Learning Sciences and sits at the intersection of design, collaboration, and computing. I design and study learning ecologies, including those supported by AI and other emerging technologies, to make high-quality collaboration and disciplinary computational practices teachable, observable, and sustainable in classrooms, teacher preparation, and professional development. Building on my longstanding work in computer-supported collaborative learning, I investigate AI agents as participants in disciplinary and collaborative activity as collaborators, mentors, peers. Three interconnected strands organize this program: (1) **high-quality collaboration**, examining how collaborative competence develops, what models of collaborative quality capture or miss, and how participation and interactions are supported in collaborative activities; (2) **AI agents as participants in learning ecologies**, investigating their designed roles and how human-AI interactions shape reasoning, learning, and agency; and (3) **disciplinary computational practices**, examining how computational thinking, data practices, and AI literacy are learned, enacted, and assessed within subjects. Across these strands, I operationalize theoretical frameworks in learning designs and analytic tools, using evidence of participants' activity to refine both the designs and theoretical accounts of practice. This program is supported by a growing portfolio of internal and external funding, anchored by an NSF CAREER award that advances this agenda through sustained design-based research in STEM learning environments.

Education

- Ph.D. in Learning and Performance Systems**, The Pennsylvania State University 2021
Emphasis: Learning Sciences
Dissertation: Shared meaning making around charged topics: Developing and operationalizing a theory for multicultural collaborative competence
- M.S. in Computer Education and Instructional Technology**, Middle East Technical University 2015
Thesis: A proposal of an instructional design model for gamified learning environments: GELD model
- B.S. in Computer Education and Instructional Technology**, Middle East Technical University 2011
- Certificate Program, Information Design**, FH JOANNEUM – University of Applied Sciences 2010

Professional Appointments

- Assistant Professor**, Texas A&M University Aug 2023 – Present
Department of Teaching, Learning, and Culture, College of Education and Human Development
- Postdoctoral Research Fellow**, University of Pennsylvania Aug 2022 – Jul 2023
Teaching, Learning, and Leadership Division, Graduate School of Education
- Postdoctoral Research Associate**, University of Connecticut Oct 2021 – Aug 2022
Department of Educational Psychology, Neag School of Education

Research Associate , The Pennsylvania State University The Collective Cognition and Design Group	Aug 2017 – Oct 2021
Graduate Research Assistant , The Pennsylvania State University Teaching and Learning with Technology (TLT)	Aug 2018 – Oct 2021
Graduate Research Assistant , The Pennsylvania State University World Campus	Aug 2015 – Jun 2018
Graduate Administrative Assistant , Middle East Technical University Instructional Technology Support Office	Dec 2012 – Jul 2015

Grants

Funded Grants

Grant Title	Agency	Role	Amount	Date
CAREER: 3AforCT: AI-Guided, Authentic, Ag-STEM Inquiry for Cultivating Computational Thinking in Rural Middle Schools	National Science Foundation (CAREER)	PI	\$1,616,223	Aug 2026 – Jul 2031
AI-Driven Professional Learning to Strengthen Pre-Service Teachers' Content Knowledge, Pedagogical Content Knowledge, and Mathematical Creativity	National Science Foundation	Co-PI	\$1,989,289	Oct 2026 – Sep 2031
AI-Enhanced Computational Thinking and Agricultural Science in Rural Schools	Catapult Grant, College of Education, Texas A&M	PI	\$30,000	2024 – 2026
Collaborative Human-AI STEM Learning Ecologies Lab (CoHALE Lab)	Vice Provost for Faculty Affairs & Dept. of Teaching, Learning, and Culture, Texas A&M	PI	\$15,000	2025 – 2027
Artificial Intelligence Mentorship for Rural Education (AIM4Rural)	Emerge Grant, College of Education, Texas A&M	PI	\$49,436	2024 – 2026
Computing and Learning Interdisciplinary Collaboration (CLIC): CLIC Research Group	Catalyst Grant, College of Education, Texas A&M	Co-PI	\$10,000	2026 – 2027
Developing and Operationalizing an Analytical Framework for Multicultural Collaborative Competence	American Educational Research Association, Division D	PI	\$5,000	2022 – 2023

Grants Under Review

Grant Title	Agency	Role	Amount	Date
CyberAI Innovation: Vigilant Horizon – Augmented Reality with Embodied AI Mentorship for Advancing Cybersecurity Competency and Trustworthy Cyberspace Workforce Development	National Science Foundation	Co-PI	\$500,000	Pending
XR-MathLeader: Immersive AI-Enabled Mathematics Instructional Rehearsal for Teachers with School-Leader Support	U.S. Department of Education	Co-PI	\$2,971,772	Pending

Non-Funded Grants (Selected)

Grant Title	Agency	Role	Amount	Date
Investigating the Persona Effects and Risks of Generative AI Chatbots on STEM Learners’ Equitable Benefits	National Science Foundation	Co-PI	\$566,269	NA
Collaborative Research: HCC: Small: Developing an Emotionally Intelligent Generative AI Chatbot (EIGA) to Empower STEM Learners	National Science Foundation	Co-PI	\$400,000	NA
Design for Belonging: Broadening Latinx Youth Participation in Computing through Student-Led Design in OctoStudio	Google	PI	\$60,000	NA
Arduino-Integrated AI-Powered Digital Science Labs	Digital Promise & Carnegie Corporation of New York	PI	\$20,000	NA
CyberThreat: Enhancing Knowledge and Interest in Cybersecurity Through an Introductory Grounding Game-Based Curriculum	National Science Foundation	Co-PI	\$2,896,457	NA

Media Coverage

Preparing Rural Students for the Future of Agriculture

Texas A&M College of Education and Human Development News

August 20, 2026

<https://education.tamu.edu/preparing-rural-students-for-the-future-of-agriculture/>

Texas A&M Researchers Explore AI to Strengthen Math Teacher Preparation

Texas A&M Stories

August 27, 2026

<https://stories.tamu.edu/news/2026/08/27/texas-am-researchers-explore-ai-to-strengthen-math-teacher-preparation/>

Teaching, Learning, and Culture Research Reaches Classrooms and Communities

Texas A&M College of Education and Human Development News

August 11, 2026

Publications

*Work with a mentored postdoctoral researcher. **Work with a mentored graduate student.

Peer-Reviewed Journal Articles

20. Kilinc, S.*, **Aldemir, T.**, Misiejuk, K., Kaliisa, R., Sabanwar, V.**, Bicer, A., Song, D. & Yalvac, B. (2026). A mixed-methods analysis of AI scaffolding patterns and student inquiry profiles in a middle-school agriculture-STEM classroom. *Computers & Education: Artificial Intelligence*, 100683. <https://doi.org/10.1016/j.caeai.2026.100683> (Impact Factor: 23.4)
19. Moon, J., **Aldemir, T.**, & Byun, G.** (2026). Design affordance diagnostic framework: Screening construct-irrelevant variance in computational thinking assessment. *Interactive Learning Environments*, 1-15. <https://doi.org/10.1080/10494820.2026.2730634> (Impact factor: 6.7)
18. **Aldemir, T.**, Bicer, A., Kilinc, S.*, Moon, J., & Kwok, M. (2026). Challenges, solutions, and PD needs for integrating AI: Insights from a two-week AI literacy module with preservice teachers. *Cogent Education*, 13(1), 2721038. <https://doi.org/10.1080/2331186X.2026.2721038> (Impact factor: 2.7)
17. Bicer, A., **Aldemir, T.**, Lee, U., Moon, J., Rambo Hernandez, K., & Sanders, M. (2026). How generative AI guided-professional development supports teachers' engagement with mathematical creativity, content knowledge, and pedagogical content knowledge. *ZDM – Mathematics Education*, 58, 1027–1039. <https://doi.org/10.1007/s11858-026-01816-1> (Impact factor: 3.8)
16. Bicer, A., **Aldemir, T.**, Gooden, M., & Colak, M. A. (2026). Examining middle school students' experiences with opportunities for mathematical creativity. *Mathematics Education Research Journal*. <https://doi.org/10.1007/s13394-026-00594-z> (Impact factor: 2)
15. **Aldemir, T.**, Yoon, S., Shim, J., & Miller, K. (2026). Unpacking relational trust mechanisms in teacher-researcher co-design partnerships: Insights from a case study. *AERA Open*, 12. <https://doi.org/10.1177/23328584261448146> (Impact factor: 2.8)
14. Bicer, A., Quiroz, F., & **Aldemir, T.** (2025). Revealing student processes in a mathematics classroom: An in-depth instrumental case study. *Learning, Culture and Social Interaction*, 52, 1-19. <https://doi.org/10.1016/j.lcsi.2025.100905> (Impact factor: 2.1)
13. **Aldemir, T.**, Kilinc, S.*, Bicer, A., Grant, P.**, Davis, T., & Sweany, N. W. (2025). Intelligent-TPACK in practice: Design and evidence from a three-week teacher preparation module. *Computers and Education Open*, 9, 100306. <https://doi.org/10.1016/j.caeo.2025.100306> (Impact factor: 5.7)
12. **Aldemir, T.**, Bicer, A., Kilinc, S.*, Moon, J., & Kwok, M. (2025). Exploring emergent AI-TPACK competencies in a two-week AI literacy module for preservice teachers. *Teaching and Teacher Education*, 168, 105231. <https://doi.org/10.1016/j.tate.2025.105231> (Impact factor: 3.9)
11. **Aldemir, T.**, Moon, J., Bicer, A., Byun, G., Perez, C. M.**, & Sabanwar, V.** (2025). How non-CS preservice teachers conceptualize and engage with computational thinking through 3D game design: Insights from an XR-enhanced maker experience. *Computers & Education: X Reality*, 6, 100099. <https://doi.org/10.1016/j.cexr.2025.100099> (Impact factor: 5.9)
10. Bicer, A., **Aldemir, T.**, Davis, T. J., & Young, J. (2025). Exploring the transferability of STEM PBL instructional principles from higher education to K-12 classrooms. *Education Sciences*, 15(1), 39. <https://doi.org/10.3390/educsci15010039> (Impact factor: 3.5)
9. Borge, M., Smith, B., & **Aldemir, T.** (2024). Using generative AI as a simulation to support higher-order thinking. *International Journal of Computer-Supported Collaborative Learning*, 19, 479-532. <https://doi.org/10.1007/s11412-024-09437-0> (Impact factor: 6.8)

8. Bicer, A., **Aldemir, T.**, Krall, G., Quiroz, F., Chamberlin, S., Nelson, J. L., Lee, Y., & Kwon, H. (2024). Exploring creativity in mathematics assessment: An analysis of standardized tests. *Thinking Skills and Creativity*, 101652. <https://doi.org/10.1016/j.tsc.2024.101652> (Impact factor: 5.3)
7. **Aldemir, T.**, Davidesco, I., Kelly, S., Glaser, N., Kyle, A. M., Montrosse-Moorhead, B., & Lane, K. (2022). Investigating students' learning experiences in a neural engineering integrated STEM high school curriculum. *Education Sciences*, 12(10), 705. <https://doi.org/10.3390/educsci12100705> (Impact factor: 3.5)
6. **Aldemir, T.**, Borge, M., & Soto, J. (2022). Shared meaning-making in online intergroup discussions around sensitive topics. *International Journal of Computer-Supported Collaborative Learning*, 17(3), 361–396. <https://doi.org/10.1007/s11412-022-09375-9> (Impact factor: 6.8)
5. Borge, M., **Aldemir, T.**, & Xia, Y. (2022). How teams learn to regulate collaborative processes with technological support. *Educational Technology Research and Development*, 70, 661-690. <https://doi.org/10.1007/s11423-022-10103-1> (Impact factor: 5.6)
4. Borge, M., Soto, J., **Aldemir, T.**, & Mena, J. (2022). Building multicultural competence by fostering collaborative skills. *Teaching of Psychology*, 49(1), 85-92. <https://doi.org/10.1177/0098628320977421> (Impact factor: 1.7)
3. Shearer, R., **Aldemir, T.**, Hitchcock, J., Resig, J., Driver, J., & Kohler, M. (2020). What students want: A vision of a future online learning experience grounded in Distance Education Theory. *American Journal of Distance Education*, 34(1), 36-52. <https://doi.org/10.1080/08923647.2019.1706019> (Impact factor: 1.3)
2. **Aldemir, T.**, Celik, B., & Kaplan, G. (2018). A qualitative investigation of student perceptions of game elements in a gamified course. *Computers in Human Behavior*, 78, 235-254. <https://doi.org/10.1016/j.chb.2017.10.001> (Impact factor: 12.2)
→ **Best Qualitative Article Award (2nd place), Division of Distance Learning, Association for Educational Communications and Technology**
1. Romero-Hall, E., **Aldemir, T.**, Colorado-Resa, J., Dickson-Deane, C., Watson, G., & Sadaf, A. (2018). Undisclosed stories of instructional design female scholars in academia. *Women's Studies International Forum*, 71, 19-28. <https://doi.org/10.1016/j.wsif.2018.09.004> (Impact factor: 2.0)

Peer-Reviewed Conference Proceedings

17. **Aldemir, T.**, Kilinc, S.*, Sabanwar, V.**, Bicer, A., & Song, D. (2026). Disciplinary computational thinking in AI-mediated problem framing: A rural Ag-STEM context. In B. K. Litts, D. DeLiema, C. Lee, S. Krist, A. Mawasi, L. Martin, & K. Kumpulainen (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences - ICLS 2026* (pp. 2985-2988). International Society of the Learning Sciences.
16. **Aldemir, T.**, Kaliisa, R., Israel-Fishelson, R., Koren, N., Oswald, E., Blikstein, P., Hochgreb, T., Fuhrmann, T., Fernandez, C., Yip, J., Druin, A., Bederson, A., Espino, D. P., Lee, S. B., & Hamilton, E. R. (2026). Bridging gaps: Participatory design as a tool for social justice in education. In B. K. Litts, D. DeLiema, C. Lee, S. Krist, A. Mawasi, L. Martin, & K. Kumpulainen (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences - ICLS 2026* (pp. 2365-2374). International Society of the Learning Sciences.
15. Sabanwar, V.**, **Aldemir, T.**, Kilinc, S.*, Bicer, A., & Song, D. (2026). How middle school students engage in computational literacies in an agricultural class. In B. K. Litts, D. DeLiema, C. Lee, S. Krist, A. Mawasi, L. Martin, & K. Kumpulainen (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences - ICLS 2026* (pp. 1657-1661). International Society of the Learning Sciences.
14. Kilinc, S.*, **Aldemir, T.**, Sabanwar, V.**, Bicer, A., & Song, D. (2026). Planners and tinkerers: A comparative case study of K-12 students' engagement with data practices in an Ag-STEM classroom. In

- B. K. Litts, D. DeLiema, C. Lee, S. Krist, A. Mawasi, L. Martin, & K. Kumpulainen (Eds.), *Proceedings of the 20th International Conference of the Learning Sciences - ICLS 2026* (pp. 726-736). International Society of the Learning Sciences.
13. Kilinc, S.*, **Aldemir, T.**, Sabanwar, V.**, Bicer, A., & Song, D. (2025). AI chatbot coaching for elevating student research. In *Proceedings of the 2025 ACM Conference on International Computing Education Research*, V. 2 (pp. 1-2).
 12. **Aldemir, T.** (2024). Navigating charged topics: Analyzing social-emotional talk in diverse online teams. In *Proceedings of the 17th International Conference of the Learning Sciences - ICLS 2024*. Buffalo, USA: International Society of the Learning Sciences.
 11. **Aldemir, T.**, Shim, J., & Yoon, S. (2024). Teachers as co-thinkers and facilitators: Unpacking teacher-student dynamics in co-design partnerships. In *Proceedings of the 17th International Conference of the Learning Sciences - ICLS 2024*. Buffalo, USA: International Society of the Learning Sciences.
 10. **Aldemir, T.**, Yoon, S., & Shim, J. (2024). Empowering voices: Promoting equitable participation in student-inclusive co-design research-practice partnerships. In *Proceedings of the 17th International Conference of the Learning Sciences - ICLS 2024*. Buffalo, USA: International Society of the Learning Sciences.
 9. **Aldemir, T.**, Yoon, S., Miller, K., & Shim, J. (2023). Teachers as co-designers: Forming equitable participation through the lens of relational trust. In *Proceedings of the 16th International Conference of the Learning Sciences - ICLS 2023*. Montréal, Canada: International Society of the Learning Sciences.
 8. **Aldemir, T.**, Borge, M., & Soto, J. (2023). How students talk about race and identity: Unpacking social-emotional talk. In *Proceedings of the 16th International Conference on Computer-Supported Collaborative Learning - CSCL 2023*. Montréal, Canada: International Society of the Learning Sciences.
 7. **Aldemir, T.**, Borge, M., & Soto, J. (2022). Perspective-taking as a tool to examine engaging in differences during shared meaning-making. In A. Weinberger, W. Chen, D. Hernández-Leo, & B. Chen (Eds.), *Proceedings of the 15th International Conference on Computer-Supported Collaborative Learning - CSCL 2022* (pp. 463-466). Hiroshima, Japan (Online Event): International Society of the Learning Sciences.
 6. Borge, M., **Aldemir, T.**, & Soto, J. (2022). Towards a framework for multicultural collaborative competence. In S. Strauß, J. Eberle, & C. Rosé (Chairs), Learning to build bridges: Promoting skills for complex collaboration across professional and cultural boundaries [Symposium]. *Proceedings of the 15th International Conference on Computer-Supported Collaborative Learning - CSCL 2022* (pp. 521–523). Hiroshima, Japan (Online Event): International Society of the Learning Sciences.
 5. **Aldemir, T.**, Borge, M., & Xia, Y. (2021). Exploration of facilitation strategies for intergroup dialogues in a CSCL context. In C. E. Hmelo-Silver, B. De Wever, & J. Oshima (Eds.), *Proceedings of the 14th International Conference on Computer-Supported Collaborative Learning - CSCL 2021* (pp. 131-138). Bochum, Germany (Online Event): International Society of the Learning Sciences.
 4. **Aldemir, T.**, & Borge, M. (2020). Unpacking collaborative sense-making: The role of reflective accuracy in collaborative process quality. In M. Gresalfi & I. S. Horn (Eds.), *The Interdisciplinarity of the Learning Sciences, 14th International Conference of the Learning Sciences (ICLS) 2020*, Volume 3 (pp. 1709-1712). Nashville, TN: International Society of the Learning Sciences. <https://repository.isls.org/handle/1/6405>
 3. Borge, M., **Aldemir, T.**, & Xia, Y. (2019). Unpacking socio-metacognitive sense-making patterns to support collaborative discourse. In K. Lund, G. P. Niccolai, E. Lavoué, C. Hmelo-Silver, G. Gweon, & M. Baker (Eds.), *A Wide Lens: Combining Embodied, Enactive, Extended, and Embedded Learning in Collaborative Settings, 13th International Conference on Computer Supported Collaborative Learning (CSCL) 2019*, Volume 1 (pp. 320-327). Lyon, France: International Society of the Learning Sciences. <https://repository.isls.org/handle/1/1584>

2. **Aldemir, T., Celik, B., & Akilli, G. K. (2016).** An evaluation study of a classroom management system: Usability and teacher experiences and evaluations. In *Proceedings from the Society for Information Technology & Teacher Education International Conference 2016* (pp. 1754-1759). Association for the Advancement of Computing in Education (AACE). <https://www.learntechlib.org/primary/p/171928/>
1. **Aldemir, T., & Kaplan, G. (2015).** Students' opinions and perceptions about a gamified online course: A qualitative study. In *Proceedings from the Association for Educational Communications and Technology Conference 2015* (pp. 1-11). Indianapolis, USA.

Book Chapters

4. Atas, A. H., Celik, B., & **Aldemir, T. (2024).** Enhancing educational assessment through GenAI: Leveraging revised Bloom's taxonomy for question generation with ChatGPT. In S. G. Turan & H. Tinmaz (Eds.), *Generative artificial intelligence applications: Holistic reflections from the educational landscape*.
3. Celik, B., Atas, A. H., & **Aldemir, T. (2022).** Sesli düşünme tekniği [Think-aloud technique] [in Turkish]. In S. Toraman-Turk (Ed.), *Sosyal bilimlerde nitel araştırmalar ve eylem araştırmaları: Yaklaşımlar, yeni arayışlar ve uygulamalar* [Qualitative research and action research in social sciences: Approaches, new trends, and applications] (Chapter 4, pp. 81-119).
2. Gregg, A., Reid, R., **Aldemir, T.**, Gray, J., Frederick, M., & Garbrick, A. (2020). Think-aloud observations to improve online course design: A case example and "how-to" guide. In M. Schmidt, A. A. Tawfik, I. Jahnke, & Y. Earnshaw (Eds.), *Learner and user experience research: An introduction for the field of learning design & technology* (Chapter 15). https://edtechbooks.org/ux/15_think_aloud_obser
1. **Aldemir, T.**, Atas, A. H., & Celik, B. (2019). A systematic design model for gamified learning environments: GELD model. In W. Tan (Ed.), *Design, motivation, and frameworks in game-based learning* (pp. 30-56). Hershey, PA: IGI Global. <https://doi.org/10.4018/978-1-5225-6026-5.ch002>

Practitioner-Focused Articles

5. Russell, S., Ramsay, C., Lonsinger, Z., Davis, S., & **Aldemir, T. (2020).** The Moral Moments Project: Where technology and compassion meet. *EDUCAUSE Review*. <https://er.educause.edu/articles/2020/6/the-moral-moments-project-where-technology-and-compassion-meet>
4. Turcotte, N., Giri, S., Vasconcelos, L., Essmiller, K., & **Aldemir, T. (2018).** The graduate student assembly: Pathways to leadership. *TechTrends*, 62(5), 427-428. <https://doi.org/10.1007/s11528-018-0322-5>
3. Anderle, J. S., **Aldemir, T.**, Resig, J., McArthur, A., Sacak, B., & DuBoff, B. (2018). Expectations and experiences of the AECT leadership internship program. *TechTrends*, 1-3. <https://doi.org/10.1007/s11528-018-0277-6>
2. Gregg, A., Reid, R., **Aldemir, T.**, Garbrick, A., Frederick, M., & Gray, J. (2018). Improving online course design with think-aloud observations: A "how-to" guide for instructional designers for conducting UX testing. *Journal of Applied Instructional Design (JAID)*, 7(2), 17-26.
1. **Aldemir, T.**, & Park, E. (2017). Student visa approved; beam me up, Scotty! Being a legal alien in American graduate school. *TechTrends*, 61(3), 210. <https://doi.org/10.1007/s11528-017-0180-6>

Research in Progress

*Work with a mentored postdoctoral researcher. **Work with a mentored graduate student.

Manuscripts Under Review

5. Awoyemi, D.** , Moon, J., **Aldemir, T.**, Ghoooreian, A.** , & Sabanwar, V.** (under review). A systematic review of empirical research on AI in K-12 STEM education (2020-2025) through a sociotechnical lens. *Educational Technology Research and Development*.
4. Albrecht, H.** , Bicer, A., & **Aldemir, T.** (under review). Rural mathematics teachers' experiences of agency, authenticity, and reflection in generative AI-mediated professional development. *Mathematics Education Research Journal*.
3. Colak, M.-A.** , Bicer, A., & **Aldemir, T.** (under review). Thinking-centered problem posing and mathematical creativity: A quasi-experimental study with preservice mathematics teachers. *Education Sciences*.
2. Bicer, A., **Aldemir, T.**, Colak, M.-A., & Albrecht, H. (under review). Fostering mathematical creativity, content knowledge, and pedagogical content knowledge through generative AI-driven creativity-directed professional development. In J. Cai (Ed.), *AI in mathematics education*. Springer. (Invited)
1. Bicer, A., **Aldemir, T.**, Sanders, M., Kopparla, M., Erdogan, N., & Lee, Y. (under review). Fostering divergent and convergent thinking through mathematics-emphasized STEM project-based learning. *Thinking Skills and Creativity*.

Articles in Preparation (Selected)

3. **Aldemir, T.**, Ordonez, L.** , Zhu, Y.** , Hou, Y.** , Atas, A. H., Bicer, A., & Song, D. (in preparation). From individual AI interactions to shared question-hypothesis revision in middle school agricultural STEM inquiry.
2. **Aldemir, T.**, Borge, M., & Pérez-Edgar, K. (in preparation). The impact of instructional support on socio-metacognitive expertise and collaborative competence in a CSCL context.
1. **Aldemir, T.**, Bicer, A., Hou, Y.** & Wang, S.** (in preparation). What makes an AI prompt productive? A lag-sequential analysis of AI moves and preservice teachers' response depth.

Presentations

*Presentations published as proceedings are listed under Peer-Reviewed Conference Proceedings. *Work with a mentored post-doctoral researcher. **Work with a mentored graduate student.*

Peer-Reviewed Presentations

36. Kilinc, S.* , **Aldemir, T.**, Sabanwar, V.** , Bicer, A., & Song, D. (2026). Computational thinking and participation in rural Ag-STEM: A comparative case study of two middle school teams [Paper presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA, United States.
35. Kilinc, S.* , **Aldemir, T.**, Sabanwar, V.** , Bicer, A., & Song, D. (2026). The AI as cognitive mentor: Scaffolding student scientific inquiry through dialogue [Paper presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA, United States.
34. Bicer, A., Quiroz, F. S., Bicer, A., **Aldemir, T.**, Gooden, M., & Melek, Z. (2026). Mathematical creativity as a unifying lens: A case study of instructional shifts in elementary classrooms through sustained professional development [Paper presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA, United States.
33. Sabanwar, V.** , **Aldemir, T.**, Kilinc, S.* , Bicer, A., & Song, D. (2026). Teacher conceptualization of student challenges and solutions for CT in a non-CS classroom [Poster presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA, United States.

32. Bicer, A., **Aldemir, T.**, Gooden, M., Bicer, A., & Quiroz, F. S. (2026). Fostering mathematical creativity, content knowledge, and pedagogical content knowledge through generative AI-driven creativity-directed professional development [Poster presentation]. American Educational Research Association Annual Meeting, Los Angeles, CA, United States.
31. Sabanwar, V**, **Aldemir, T.**, Kilinc, S.*, Bicer, A., & Song, D. (2025). Exploring computational thinking (CT) skills in the era of GenAI in K–12 education. Paper presented at the 2025 AECT International Convention, Las Vegas, NV, United States.
30. Kilinc, S.*, **Aldemir, T.**, Sabanwar, V**, Bicer, A., & Song, D. (2025). Enhancing research questions and hypothesis development through AI chatbot assistance in rural agricultural education. Paper presented at the 2025 AECT International Convention, Las Vegas, NV, United States.
29. Bicer, A., & **Aldemir, T.** (2025). Fostering mathematical creativity through AI-driven professional development. Paper presented at the 2025 School Science and Mathematics Association Convention, Fort Worth, TX.
28. Bicer, A., **Aldemir, T.**, Krall, G., Quiroz, F., Chamberlin, S., Nelson, J., Lee, Y., & Kwon, H. (2025). Creativity in mathematics assessment. 9th ICMI-East Asia Regional Conference on Mathematics Education, Jung-gu, Seoul, Korea.
27. Bicer, A., Bicer, A., **Aldemir, T.**, Gooden, M., Sanders, M., Lee, Y., Perihan, C., & Quiroz, F. (2025). Enhancing PSTs' mathematical creativity via STEM PBL: A mixed-methods study. Paper presented at the annual meeting of the American Educational Research Association (AERA), Denver, CO.
26. Bicer, A., **Aldemir, T.**, Gooden, M., Erdogan, N., Sanders, M., & Quiroz, F. (2025). STEM PBL in the context of mathematical creativity. Paper presented at the annual meeting of the American Educational Research Association (AERA), Denver, CO.
25. Bicer, A., Gooden, M., Sanders, M., **Aldemir, T.**, Kopparla, M., Lee, Y., & Calabrese, J. (2025). Problem-posing as a means for developing pre-service teachers' mathematical creativity. Paper presented at the annual meeting of the American Educational Research Association (AERA), Denver, CO.
24. **Aldemir, T.**, Shim, J., & Yoon, S. (2024). Tokenistic or genuine: Engaging students as co-designers of a curriculum unit. Roundtable session presented at the American Educational Research Association (AERA) Convention, Philadelphia, USA.
23. Glaser, N., Center, M., Jensen, L., Davidson, A., & **Aldemir, T.** (2023). Monsters: Redefining monstrosity: Compassion and duty in 'Spiritfarer'. Presented at the MultiPlay Network Video Game Monster Conference. Virtual Convention.
22. **Aldemir, T.**, & Borge, M. (2023, April 13–16). Conceptualizing quality of intergroup dialogue in CSCL: Analysis of social and emotional talk. Full paper session presented at the American Educational Research Association (AERA) Convention, Chicago, USA.
21. **Aldemir, T.**, & Borge, M. (2022, October 24–28). Leveraging a theoretically informed technological intervention to support multicultural collaborative competence. Concurrent session presented at the Association of Educational Communications & Technology (AECT), Las Vegas, NV.
→ **McJulien Scholar Best Paper Award**
20. **Aldemir, T.**, Glaser, N., & Davidesco, I. (2022, October 24–28). Exploring how user-centered approaches and empathy informed the co-design of a neural engineering curriculum. Concurrent session presented at the Association of Educational Communications & Technology (AECT), Las Vegas, NV.
19. Glaser, N., Herman, K., & **Aldemir, T.** (2022, October 24–28). Artificial intelligence and computer vision: Opportunities and challenges for instructional design and evaluation. Roundtable session presented at the Association of Educational Communications & Technology (AECT), Las Vegas, NV.

18. **Aldemir, T.**, Borge, M., & Soto, J. (2021, April 8–12). Sense-making patterns in online intergroup dialogues around politically charged topics. Poster session presented at the American Educational Research Association (AERA) Convention. Virtual Convention.
17. **Aldemir, T.**, & Borge, M. (2020, November 2–7). Learning to monitor and regulate socio-emotional interactions when discussing politically charged topics. Concurrent session presented at the Association of Educational Communications & Technology (AECT). Virtual Convention.
16. Rogers, S. E., Dickson-Deane, C., Liu, J. C., Asino, T., Subramony, D. P., Heggart, K., & **Aldemir, T.** (2020, November 2–7). What is our responsibility to those prevented from learning by governmental policy decisions? Presidential Panel presented at the Association of Educational Communications & Technology (AECT). Virtual Convention.
15. **Aldemir, T.**, & Borge, M. (2019, April 5–9). Socio-metacognitive processes in computer-supported collaborative learning. Full paper session presented at the American Educational Research Association (AERA) Convention, Toronto, Canada.
14. **Aldemir, T.** (2018, October 23–27). Re-considering triangulation: A critical realist approach to the mixed methods research praxis in social science. Concurrent session presented at the Association of Educational Communications & Technology (AECT) Convention, Kansas City, MO, USA.
13. Resig, J., DuBoff, B., **Aldemir, T.**, & Anderle, J. (2018, October 23–27). AECT 2017 leadership intern class project: Facilitating organizational advancement through membership data analysis. Concurrent session presented at the Association of Educational Communications & Technology (AECT) Convention, Kansas City, MO, USA.
12. **Aldemir, T.**, Atas, A. H., & Celik, B. (2018, October 23–27). Proposal of an instructional design model for gamified learning environments: GELD model. Concurrent session presented at the Association of Educational Communications & Technology (AECT) Convention, Kansas City, MO, USA.
11. Shearer, R., **Aldemir, T.**, & Bourland, J. (2017, November 7–11). Future learning experience for distance education. Concurrent session presented at the Association of Educational Communications & Technology (AECT) Convention, Jacksonville, FL, USA.
10. Zhongrui, Y., Neumann, K., **Aldemir, T.**, Arrington, L., & Bae, H. (2017, November 7–11). Student organizations: Experiences and reflections from student leaders. Panel discussion presented at the Association of Educational Communications & Technology (AECT) Convention, Jacksonville, FL, USA.
9. Romero, E. J., Colorado, Z., Watson, G., Sadaf, A., **Aldemir, T.**, & Dickson-Deane, C. (2017, November 7–11). Instructional design research women's caucus. Panel discussion presented at the Annual Conference of the Association for Educational Communications and Technology (AECT) Convention, Jacksonville, FL, USA.
8. Gregg, A., Gray, J., & **Aldemir, T.** (2017, July). Leveraging User Experience (UX) research to improve Canvas course design. The Pennsylvania State University Learning Design Summer Camp, State College, PA, USA.
7. Shearer, R., Gregg, A., & **Aldemir, T.** (2016, October 17–21). Theory, practice, and future learning experience. Concurrent session presented at the Association of Educational Communications & Technology (AECT) Convention, Las Vegas, NV, USA.
6. Celik, B., & **Aldemir, T.** (2016, October 17–21). Student perceptions of usability, enjoyment, and vocabulary learning: A case of Vocabulary.com. Concurrent session presented at the Association of Educational Communications & Technology (AECT), Las Vegas, NV, USA.
5. **Aldemir, T.**, Celik, B., & Kaplan, G. (2016, March 21–25). An evaluation study of a classroom management system: Usability and teacher experiences and evaluations. Concurrent session presented

at the 27th Annual Conference of the Society for Information Technology and Teacher Education (SITE), Savannah, GA, USA.

4. **Aldemir, T.**, & Kaplan, G. (2015, November 3–7). Students' opinions and perceptions about a gamified online course: A qualitative study. Concurrent session presented at the Association of Educational Communications & Technology (AECT) Convention, Indianapolis, IN, USA.
3. **Aldemir, T.**, Gurel-Koybasi, N., Teker, M., & Cagiltay, K. (2015, February 4–5). Migration to a new learning management system: ODTUClass experience (in Turkish). Concurrent session presented at the Akademik Bilisim Conference, Eskisehir, Turkey.
2. **Aldemir, T.**, & Kaplan, G. (2014, September 27–28). A new approach to improve students' motivation and achievement in ESL writing: Project-based digital storytelling. Concurrent session presented at the 8th International Computer and Instructional Technologies Symposium, Edirne, Turkey.
1. Atas, A., **Aldemir, T.**, & Delialioglu, O. (2014, September 27–28). Acceptance of a mobile question-answer system: Application of the UTAUT model. Concurrent session presented at the 8th International Computer and Instructional Technologies Symposium, Edirne, Turkey.

Invited Lectures and Presentations

6. **Aldemir, T.** (2023, November 21). Fostering diverse cultural identities in the classroom [Audio podcast episode]. In *Podcast for the Center for Engaged Learning*, Elon University. <https://www.centerforengagedlearning.org/fostering-diverse-cultural-identities/>
5. Glaser, N., Earnshaw, Y., **Aldemir, T.**, & Zoubi, D. (2022, June). Navigating the job market: Insight from the inside: How to land your dream job in academia or industry. Panel discussion presented at the 8th International Conference of the Immersive Learning Research Network, Online Event.
4. **Aldemir, T.** (2022, March). Wise integration of technology into teaching and learning environments. Guest lecture, EPSY 5530: Theories of Learning, Cognition and Instruction, Neag School of Education, University of Connecticut, Storrs, CT.
3. Gregg, A., Reid, R., **Aldemir, T.**, Garbrick, A., & Gray, J. (2022, April). Think-aloud methods for LXD: Just-in-time & systematic methods to improve course designs. Webinar presented at the AECT Design and Development Division LXD Webinar Series.
2. **Aldemir, T.**, Cadieux-Boulden, D., Gyabak, K., Mao, Z., & Sacak, B. (2018, April). The future of design & technology. Panel discussion presented at the AERA Convention, Design and Technology SIG Business Meeting, New York, NY, USA. (Speakers in alphabetical order)
1. **Aldemir, T.**, & Kaplan, G. (2015, May). Effective presentation techniques and Web 2.0 presentation tools [Series of seminars]. Türkiye Merkez Bankası (Central Bank of Turkey), Ankara, Turkey.

Teaching

Teaching Certificate

Online Teaching Certificate, The Pennsylvania State University, World Campus

June 2018

Courses Taught

Course	Format	Level	Term	Enroll.	Eval.	Role
Texas A&M University						
LDTC 649: Teaching and Learning with AI	Async.	Master's	Fall 2025	13	4.80/5	Instructor

Course	Format	Level	Term	Enroll.	Eval.	Role
LDTC 649: Teaching and Learning with AI	F2F	PhD	Spring 2026	5	4.75/5	Instructor
EDCI 365: Technology in Classrooms	F2F	UG	Fall 2023	28	4.00/5	Instructor
				29	4.86/5	
EDCI 365: Technology in Classrooms	F2F	UG	Fall 2024	29	4.18/5	Instructor
EDCI 365: Technology in Classrooms	F2F	UG	Spring 2025	30	4.50/5	Instructor
University of Pennsylvania						
EDUC 6139: Design of Learning Environments	F2F	Master's / PhD	Spring 2023	7	3.86/4	Instructor
The Pennsylvania State University						
LDT 100: World Technologies and Learning	Online / F2F	UG	Spring 2017; Fall 2020	30/section	–	TA
PSYCH 432: Multicultural Psychology	F2F	UG	Fall 2017; Fall 2019; Fall 2020	30/section	–	TA
Centre County Correctional Facility, Bellefonte, PA (Restorative Justice Initiative)						
Entrepreneurship	F2F	–	Spring 2019	12	–	TA
Middle East Technical University, Ankara, Turkey						
CEIT 323: Instructional Technology and Material Development	Blended	UG	Fall 2014; Spring 2015	35; 33	–	Co-Instr.
Professional Development Program on Instructional Technologies	F2F	Faculty / Staff	Fall 2013 – Spring 2015	100+	–	Trainer
Bilim and Arı Private Primary Schools, Ankara, Turkey						
Computer Science, 4th Grade	F2F	Grade 4	Fall 2010; Spring 2011	30; 28	–	Intern

Abbreviations: F2F = face-to-face; Async. = asynchronous; UG = undergraduate; TA = teaching assistant; Co-Instr. = co-instructor.

Curriculum Design and Development

LDTC 649: Teaching and Learning with AI , Texas A&M University	2025 – 2026
– New Course Design; Asynchronous; Master's Level (2025)	
– Course Redesign; Face-to-Face; Doctoral Level (2026)	
Sustainable Agriculture Curriculum Unit , Texas A&M University & Two Rural School Districts	2024 – 2026
– New Curriculum Unit Design; Grades 7–8	
– Co-Design with Rural Agriculture Teachers; Implemented with Approximately 240 Students	
AI Literacy Module for Teacher Preparation , Texas A&M University	2024 – 2026
– New Module Design; Two-Week Module; Preservice Teachers	

- Computational Thinking through 3D Game Design**, Texas A&M University 2024 – 2025
 – Module Redesign; Three-Week XR-Enhanced Maker Module; Preservice Teachers
- CEIT 323: Instructional Technology and Material Development**, Middle East Technical University 2014 – 2015
 – Course Redesign; Preservice Teachers
- Professional Development Program on Instructional Technologies**, Middle East Technical University 2013 – 2015
 – New Professional Development Program Design; University Faculty and Staff

AI-Supported Learning Systems and Agent Development

- AI Peer Mentor and Agent Design Framework for Disciplinary Computational Practices in Problem-Based Inquiry** 2025 – Present
 – Design Framework and Technical Infrastructure for AI Agents Supporting Disciplinary Computational Practices in Problem-Based Inquiry
 – Persona-Based Pedagogical Framework in Which Agents Assume Distinct Roles Across Phases of the Inquiry Unit; Prototyped in Google AI Studio and Deployed in Middle School Classrooms
- Conversational AI Agents for AI-Guided Teacher Professional Learning** 2025 – Present
 – Co-Developed Conversational AI Agents in Hugging Face for Creativity-Directed Professional Learning with Preservice Mathematics Teachers
 – Agent Instructions Encode Module-Specific Pedagogical Specifications Authored by the Subject-Matter Expert (PI) and Guide Teachers Through Module Tasks

Advising and Mentoring

Graduate Advising and Committee Service

Name	Year	Level / Department	Role	Status
Yaodong Xu	2026 – Present	PhD, Teaching, Learning and Culture	Chair	Pre-Candidacy
Yuxuan Hou	2026 – Present	Master's, Teaching, Learning and Culture	Chair	In-Progress
Lauren Ordonez	2025 – Present	PhD, Teaching, Learning and Culture	Chair	Pre-Candidacy
Carlos Manrique Perez	2024 – Present	PhD, Teaching, Learning and Culture	Chair	Pre-Candidacy
Weiqi Liu	2025 – Present	PhD, Educational Psychology	Member	Pre-Candidacy
Patricia Grant	2024 – Present	PhD, Educational Psychology	Member	Pre-Candidacy
Nkemdi Anyiam	2025 – Present	PhD, Computer Science & Engineering	Member	Pre-Candidacy
Seneca Hart	2024 – Present	EdD, Teaching, Learning and Culture	Member	Candidate

Name	Year	Level / Department	Role	Status
Sami El Ahmadie	2023 – Present	PhD, Educational Human Resource Development	Member	Candidate
Brittney Hawkins	2022 – Graduated	EdD, Teaching, Learning and Culture	Co-Chair	Graduated
Shweta Kailani	2023 – Graduated	PhD, Educational Psychology	Co-Chair	Graduated
Jayant Surendra Palkar	2023 – Graduated	PhD, Educational Psychology	Member	Graduated

Mentoring

CoHALE Lab team	One postdoctoral researcher (2025–2026), two doctoral students, one master’s student, and two undergraduate researchers.
Areas of mentoring	Establishing and sustaining school partnerships; designing and implementing interventions in classrooms; collecting and analyzing multimodal data; writing and disseminating scholarship.

Honors, Awards, and Fellowships

Awards and Recognition

Reviewer of the Year , <i>Journal of the Learning Sciences</i>	2026
McJulien Scholar Best Paper Award , Culture, Learning, and Technology Division, AECT	2022
ISLS Early Career Symposium , International Society of the Learning Sciences (\$4,000)	2024
NSF ECR Funding PI Meeting, Prospective PI (Invited) , NSF ECR Hub (\$2,000)	2024
Simon Initiative LearnLab Summer School Award , Carnegie Mellon University, LearnLab (\$250)	2021
Annual Meeting Registration Award , International Society of the Learning Sciences (\$125)	2021
Innovative Practices: CSCL in a Time of Crisis, “Learning Together Online” , ISLS & Network of Academic Programs in the Learning Sciences (NAPLeS) (\$800)	2020
Outstanding Service Award , DDL Communication Officer, Division of Distance Learning, AECT	2020
Early Career Symposium , Division of Research, Theory and Development, AECT	2020
Best Qualitative Article Award (2nd place) , Division of Distance Learning, AECT	2018
McJulien Graduate Student Award , Culture, Learning, and Technology Division, AECT (\$500)	2018
Annual Meeting Registration Award , Division of Organizational Training and Performance, AECT (\$165)	2018
Outstanding Service Award , OTL SIG Newsletter Content Editor, Online Teaching and Learning SIG, AERA	2018
Lee W. Cochran Internship , AECT Leadership Development Internship Program (\$1,500)	2017
Conference Internship , Pennsylvania Educational Technology Expo and Conference (PETE&C) (\$1,000)	2016

Fellowships and Scholarships

Miriam E. Gray Scholarship , College of Education, The Pennsylvania State University (\$2,400 per year)	2016 – 2020
Graduate Fellowship , World Campus, The Pennsylvania State University (\$38,000 per year)	2015 – 2018
“Pervasive Networks and Service Infrastructures” Programme , Technological Educational Institute of Crete, Greece; funded by the State Scholarships Foundation of the Greek Ministry of Education under the ERASMUS Lifelong Learning framework (€2,500)	July 2013
Erasmus Student Exchange Program Fellowship , European Union (€2,000)	Spring 2010

Professional Service

Editorial Service

Guest Editor , European Journal of STEM Education	Sep 2025 – Jun 2026
Special Issue: <i>AI and STEM: Exploring the Impact of Artificial Intelligence on Science, Technology, Engineering, and Mathematics Education</i>	

Grant Review

Improving Undergraduate STEM Education (IUSE: EDU), National Science Foundation	2026
Tri-Agency Interdisciplinary - Insight Grant and Discovery Horizons, Canadian Institutes of Health Research	2026
Innovative Technology Experiences for Students and Teachers (ITEST), National Science Foundation	2024
Computer Science for All (CSforAll), National Science Foundation	2024
Center for Online Innovation in Learning (COIL) Seed Grants, The Pennsylvania State University	2016 - 2017

Book Review

<i>Sensemaking in the Age of AI: Preserving Scientific Thinking in an Era of Instant Answers</i> Cambridge University Press & Assessment	2026
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Journal Review (Selected)

British Journal of Educational Technology	2026 – Present
Computer Science Education	2026 – Present
Educational Technology Research and Development	2026 – Present
Instructional Science	2026 – Present
Interactive Learning Environments	2026 – Present
International Journal of Computer-Supported Collaborative Learning	2024 – Present
Journal of the Learning Sciences → Reviewer of the Year Award, 2026	2024 – Present
Computers in Human Behavior	2018 – Present

Conference Review

International Conference of the Learning Sciences (ICLS)	2017 – Present
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International Conference on Computer-Supported Collaborative Learning (CSCL)	2017 – Present
American Educational Research Association (AERA) Annual Meeting	2016 – Present
Association for Educational Communications and Technology (AECT) Annual Meeting	2016 – Present

Professional Organization Leadership & Committee Service

International Society of the Learning Sciences (ISLS)

Co-Chair, Membership Committee	2024 – Present
Member, Annual Conference Planning Committee	2022 – 2024
Member Outreach Officer, Learning Sciences Students Association	2021 – 2023
Communication Officer, Learning Sciences Students Association	2019 – 2021
Co-Founder, Learning Sciences Students Association	2019

American Educational Research Association (AERA)

Newsletter Content Editor (Bi-Monthly), Online Teaching and Learning SIG (OTL)	2017 – 2019
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Association for Educational Communications and Technology (AECT)

VP of Awards, Division of Culture, Learning, and Technology (CLT)	2021 – 2024
VP of Finance & Development, Division of Culture, Learning, and Technology (CLT)	2019 – 2021
Communication Officer, Division of Distance Learning (DDL)	2019 – 2020
GSA Board Representative, Graduate Student Assembly (GSA)	2018 – 2019
Member, Social Media Task Force	2018
Member, Leadership Committee	2017 – 2019
President, Graduate Student Assembly (GSA)	2017 – 2018
Division of Design and Development Representative, Graduate Student Assembly (GSA)	2016 – 2017

Conference & Event Organizing and Session Chairing

International Society of the Learning Sciences (ISLS)

Co-Organizer, “Chats and Coffee with Professors” Mentorship Series	2021 – Present
Co-Planner & Co-Moderator, ILSSA Sessions, ISLS Annual Meeting	2020 – 2022
Co-Planner & Co-Moderator, First-Timers Sessions, ISLS Annual Meeting	2020 – 2022

American Educational Research Association (AERA)

Session Chair, Annual Meeting (Division C: Learning & Instruction; Learning Sciences SIG)	2017 – Present
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Association for Educational Communications and Technology (AECT)

Session Chair, Annual Meeting (Divisions of CLT, DDL)	2017 – Present
Planner & Co-Moderator, Graduate Student Assembly (GSA) Sessions, Annual Meeting	2018
Co-Moderator & Co-Coordinator, Design & Development Reads Program, Division of Design and Development	2017

University & Departmental Service

Texas A&M University

Chair, Undergraduate Scholarship Committee, College of Education and Human Development 2023 – Present

Member, Faculty Working Group, Department of Teaching, Learning, and Culture 2024 – 2025

Organizer, Guest Speaker Academic Talk, College of Education and Human Development 2023

The Pennsylvania State University

Chair, Graduate Student Committee, Learning, Design, and Technology Program 2017

Co-Planner & Co-Moderator, First Friday Speaker Series, Learning, Design, and Technology Program 2017

Professional Development

Faculty Success Program (FSP), National Center for Faculty Development & Diversity (NCFDD) Sep 2024 – Jan 2025

EMG Data Collection and Analysis Workshop, Cognitive Sciences Shared Electrophysiology Resource Laboratory, University of Connecticut Feb – Mar 2022

Simon Initiative Summer School: Computational Models of Learning, LearnLab, Carnegie Mellon University July 2021

Machine Learning with LightSIDE Workshop, Elijah Mayfield and Carolyn P. Rosé, Carnegie Mellon University September 2019

Graduate Mentoring Program, Design and Technology SIG, American Educational Research Association April 2018

Professional Affiliations

Association for Computing Machinery (ACM) 2024 – Present

International Society of the Learning Sciences (ISLS) 2017 – Present

American Educational Research Association (AERA) 2016 – Present

Association for Educational Communications and Technology (AECT) 2015 – Present

Languages

Turkish (native); English (full professional proficiency)

Last updated: September 28, 2026